
Assignment 1: The CNN Challenge

ITCS 6169/8169 – Computer Vision

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Code & checkpoint: <https://github.com/sasukepn1999/itcs6169-assignment1> **Test accuracy:** $93.1 \pm 0.8\%$ **Selected-model val. accuracy:** $93.7 \pm 0.5\%$

1 Final Result

Final model (Sec. 4): ImageNet-pretrained EfficientNet-B0, fine-tuned with FixMatch on unlabelled images. We use three sets of images:

Validation set (480 labelled images held out from the provided training set, seeded split as in the starter). Every design choice is made here, with 3 seeds per configuration; we report best-epoch accuracy and the mean of the last 10 epochs, and select on the latter (the maximum of a noisy curve is optimistic). The final model’s supervised part (Exp. 9: trained on the other 1,920 images, without FixMatch) reaches $93.7 \pm 0.5\%$.

Test set (the 400 labelled test images provided with the assignment). Evaluated only for final models. Our test accuracy, $93.1 \pm 0.8\%$ (seeds 93.5 / 92.0 / 93.8), comes from the final recipe trained on train+val (2,400 images) with FixMatch on test2; these models never see the test images (neither labels nor pixels). For reference: best model from scratch $81.8 \pm 1.1\%$, starter 42.3%.

test2 (400 unlabelled images): only used as unlabelled FixMatch data (no labels, no accuracy).

Checkpoints (Google Drive): two models with the same recipe. The *clean-test* models (train+val) give the test accuracy above; the *submitted* model, also trained on the labelled test set, produces the test2 predictions and has no score (no labelled data left).

2 Data Analysis

The dataset is perfectly class-balanced (150 training and 25 test images per class), so no imbalance handling is needed. Images are far larger than 64^2 (median 256^2). **Image format leaks the label** (also on test): eight classes are exactly 256×256 , Suburb 330×220 , and *Flower is the only RGB class* and the largest, so colour alone could identify it (Sec. 5).

3 Baseline and Experimental Journey

Starter baseline. TNet (one 3×3 conv, 4×4 max-pool, linear classifier; 58K parameters; 64^2 gray; 20 epochs) reaches 49.8% validation and 42.3% test, reproduced exactly by our code. It overfits severely (training loss 0.12; validation loss lowest at epochs 5–7) and is weakest on indoor scenes (30–35%).

Exp. 1: augmentation and training length. Added one at a time, translation (reflect-pad + random crop) helps most and nearly closes the train/val gap (final loss 1.33/1.49 vs. 0.15/2.18); random-resized crops (RRC) and flips help less; rotation, jitter and erasing do not. Combining the three and training longer helps most, with little gain beyond 200 epochs.

Exp. 2: learning-rate schedule. Per-iteration decay (step, cosine, warm-up + cosine, one-cycle) beats constant lr at 200 epochs (63.4–64.1 vs. 62.8%; exponential decay, too early, 60.9%), and cosine tolerates a larger peak lr (5×10^{-3} : 65.2%, above constant lr at 600 epochs).

Exp. 3: resolution, revisited. With translation scaled to $s/8$, 200^2 is best (68.3%) and 256^2 is worst (61.9%), reversing a sweep made before fixing overfitting (TNet’s classifier also grows with s).

Exp. 4: deeper CNNs (from scratch). TNet’s training loss stays near 1.0, so it is capacity-limited. Seven torchvision CNNs from random init (better of two peak lrs kept) reach 81.6–83.5%, except ConvNeXt-Tiny (74.5%). **EfficientNet-B0** (4.0M params) ties ResNet-50 (23.5M) at 83.5% with 6× fewer parameters, so we keep it; it overfits again (training loss ≈ 0.01).

Exp. 6–7: self-supervised auxiliary task. An SSL loss (rotation, SimCLR, SimSiam or MAE-style masked modelling) on the training images does not reduce overfitting (rotation: within noise at 2× cost; others: -2 to -10), nor on the unlabelled validation images (80.5–82.5% test vs. 81.8%).

Exp. 8: more labels and FixMatch. Training on all 2,400 train+val images raises test accuracy to $84.8 \pm 1.2\%$ (+3.1); FixMatch on the unlabelled test images (threshold 0.95, $\lambda_u = 1$, untuned) then gives $86.9 \pm 1.4\%$ (+2.1, transductive, ≈ 1.5 std). Its pseudo-labels end up covering 98% of the images, but their accuracy drops from $\approx 95\%$ to 86–89% (confirmation bias).

Exp. 9: ImageNet initialisation. Starting from ImageNet-1k weights (all layers fine-tuned) is the largest single gain: **$93.7 \pm 0.5\%$** at peak lr 10^{-4} vs. 83.5% from scratch (+10.2); larger lrs are worse (92.5 / 90.6%).

Table 1: Experimental journey (val. accuracy %, mean $\pm$ std, 3 seeds); each row builds on the one above. *Best ep.*: max over epochs; *Last-10*: mean of the final 10 epochs (selection metric).

Experiment	Best ep.	Last-10	Observation
Starter TNet (64 ² , gray, 20 ep.)	49.6 $\pm$ 0.8	46.9 $\pm$ 1.4	Overfits after epoch ~ 7
+ translation (Exp. 1)	54.0 $\pm$ 1.6	51.0 $\pm$ 1.2	Rotation/jitter/erasing: no gain
+ flip + RRC, 100 epochs (Exp. 1b)	63.7 $\pm$ 1.5	60.5 $\pm$ 1.1	Translation alone: 58.4 (overfits again)
+ cosine lr 5×10^{-3} , 200 ep. (Exp. 2)	66.3 $\pm$ 1.5	65.2 $\pm$ 0.9	Constant lr, 600 ep.: 64.3
+ input 200 ² (Exp. 3)	69.7 $\pm$ 1.4	68.3 $\pm$ 0.2	224 ² : 67.0; 256 ² : 61.9
+ EfficientNet-B0 (Exp. 4)	85.0 $\pm$ 0.2	83.5 $\pm$ 0.5	ResNet-50 83.5; ConvNeXt-T 74.5
+ rotation SSL, $\lambda = 1$ (Exp. 6)	85.8 $\pm$ 1.0	84.4 $\pm$ 0.9	Within noise, 2× cost
+ ImageNet init., lr 10^{-4} (Exp. 9)	94.7 $\pm$ 0.7	93.7$\pm$0.5	lr 10^{-3} : 92.5; 5×10^{-3} : 90.6

4 Our Secret Recipe

EfficientNet-B0 (4.0M parameters) initialised from **ImageNet-1k** (torchvision IMAGENET1K_V1), all layers fine-tuned, new 16-way head; 200×200 grayscale replicated to 3 channels, ImageNet normalisation; random-resized crop (scale 0.5–1) + reflect-pad translation ($s/8$) + horizontal flip; Adam, peak lr 10^{-4} , per-iteration cosine decay to 0; batch 64; 200 epochs; cross-entropy; **FixMatch** on the unlabelled test2 images (pseudo-label threshold 0.95, $\lambda_u = 1$, strong view: RandAugment(2, 10) + Cutout; 64 + 64 images per step); final-epoch weights. **What mattered** (Table 1): ImageNet initialisation (+10.2), a deeper CNN (+15.2 over TNet), augmentation with longer training (+15.9 for TNet), a cosine schedule (+2.4), and, on test (Exp. 8), more labels (+3.1) and FixMatch (+2.1). Grayscale inputs avoid a colour shortcut (Sec. 5).

5 Failure Analysis

Keeping colour for the colour images (Exp. 5). *Tried*: keep the originally-RGB images (only Flower) in colour. *Why*: colour is genuine flower information. *What happened*: validation accuracy fell (EfficientNet-B0 83.5 $\rightarrow$ 80.9%, ResNet-50 83.5 $\rightarrow$ 82.8%), and Flower accuracy did not improve (91.4% gray vs. 90.3 / 84.9%). Shown in gray at test time, Flower images are recognised only 25.8 / 18.3% of the time (gray models: unchanged), and gray images are almost never called Flower (0.07 vs. 0.5%): the models learned “colour $\Rightarrow$ Flower”, which would pass our test set (Flower is again the only colour class) but fail on gray flowers. *Learned*: a nuisance attribute perfectly correlated with one label must be removed or probed; accuracy alone can hide a shortcut.

6 AI + Human

AI help: Claude Code help implement all the experiments and draft report. *Human judgement*: I am mainly responsible for designing experiments and hypothesis based on what I observe on the results (e.g., loss curve) and my experience. Details in AI_USAGE.md.